

Optimizing Metal-Organic Framework Monoliths for Carbon Dioxide Adsorption Using Computer-Aided Design and Finite Element Analysis

Background

Problem: Carbon dioxide traps heat in atmosphere, causing rapid climate change

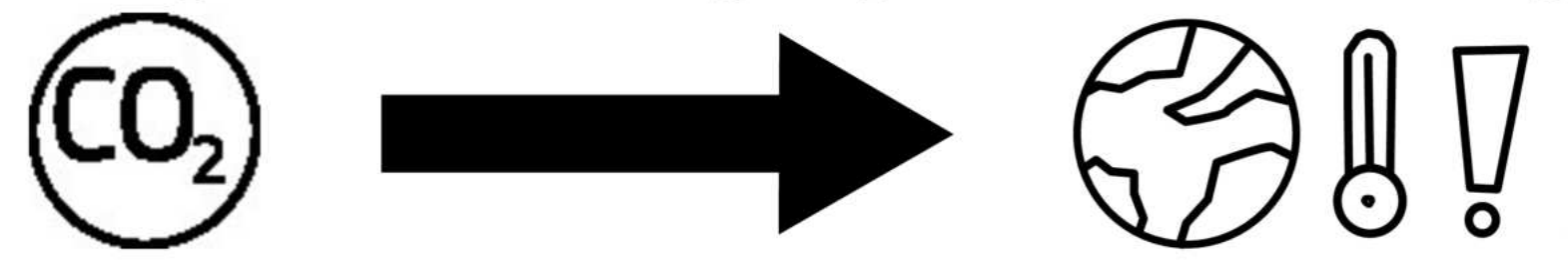


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Solution: Carbon capture

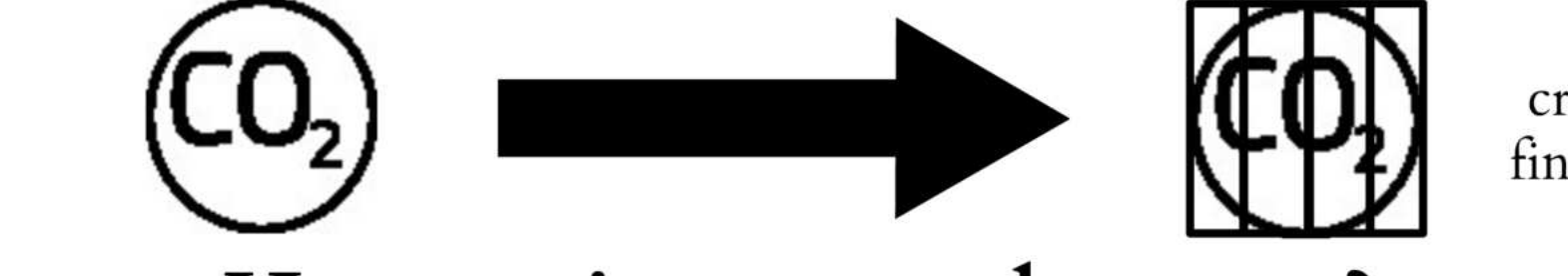


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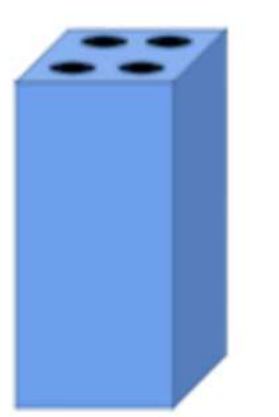
How to improve the cage?

Adsorption-based carbon capture: use of adsorbents to separate CO₂ from other gases with minimal energy usage.

Metal-organic frameworks (MOFs): make for effective adsorbents due to their cage-like shape.

Previous Studies

Monoliths: Solid adsorption reactors (Stampi-Bombelli et al., 2024)



Activated carbon monoliths (Wang et al., 2021)

Graphic created by finalist with Google Slides

- Insufficient research conducted using:
- Computer-Aided Design (CAD)
 - Finite-Element Modeling (FEM)
 - Faster discovery.
 - MOF-based monoliths
 - Economic analysis
 - Cost-friendly simulation methods

Research Goal

Optimize a MOF monolith in terms of mass transfer, heat transfer, structural stability, and adsorption cost using CAD/FEM.

Methods (Tools)



freecad.org

- Finite element modeling/CAD software
- 20 monoliths (random pattern origin)
 - Square, circle, hexagon, triangle shapes
 - 0.25, 0.5, 1, 1.5, 2 mm pore sizes

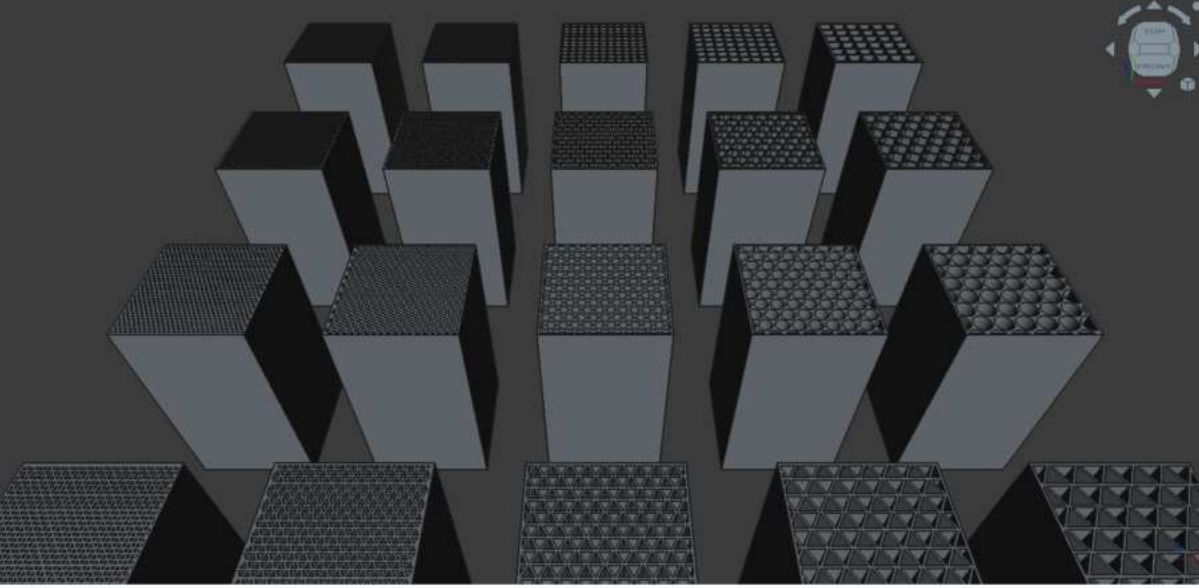


Figure 1 - View of all 20 monolith meshes

Methods (Procedure)

Mass Transfer

- **Adsorption capacity:** amount of CO₂ that can be held in an MOF (in mmol/g)
- Maximize entry points in monolith (surface area to volume ratio)

Material Adsorption Cost

HKUST-1 + MOF-5 Adsorption/Heat/Cost Data → Excel → Cost per mol CO₂ adsorbed

Heat Transfer

Figure created by finalist with ParaView

Simulate single monolith channel

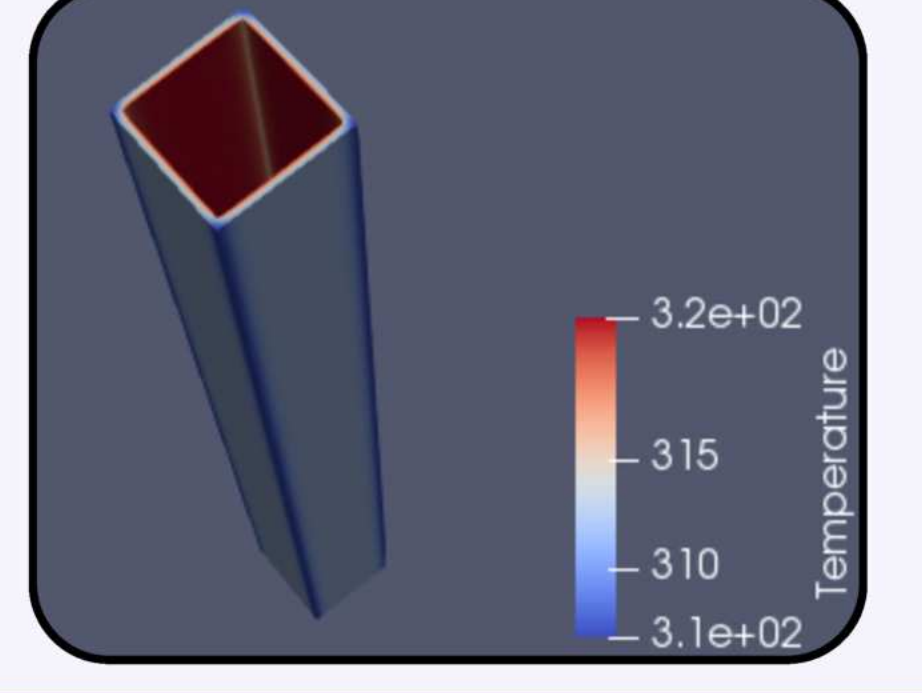
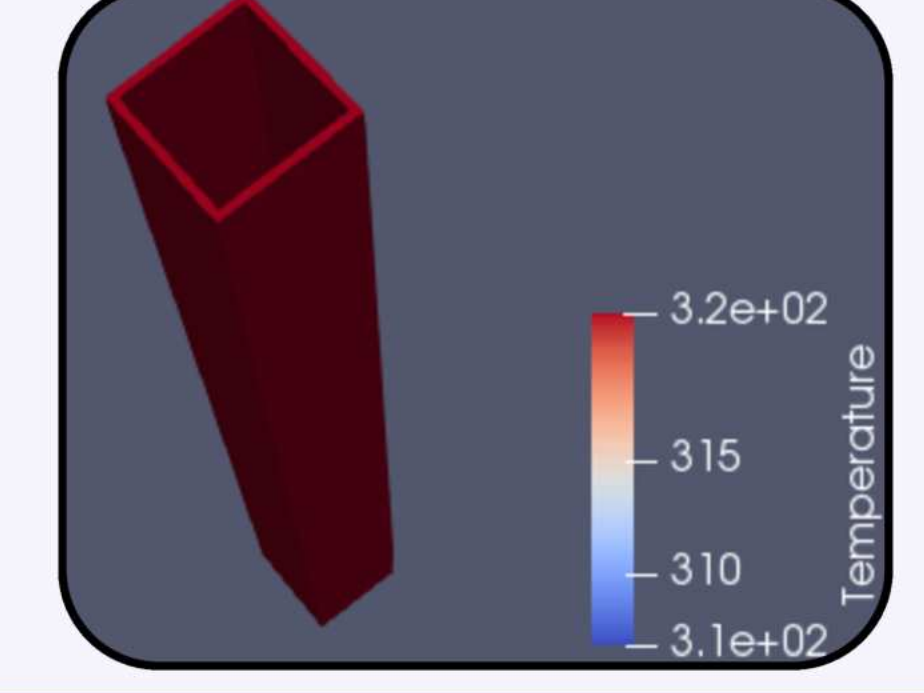


Figure created by finalist with ParaView

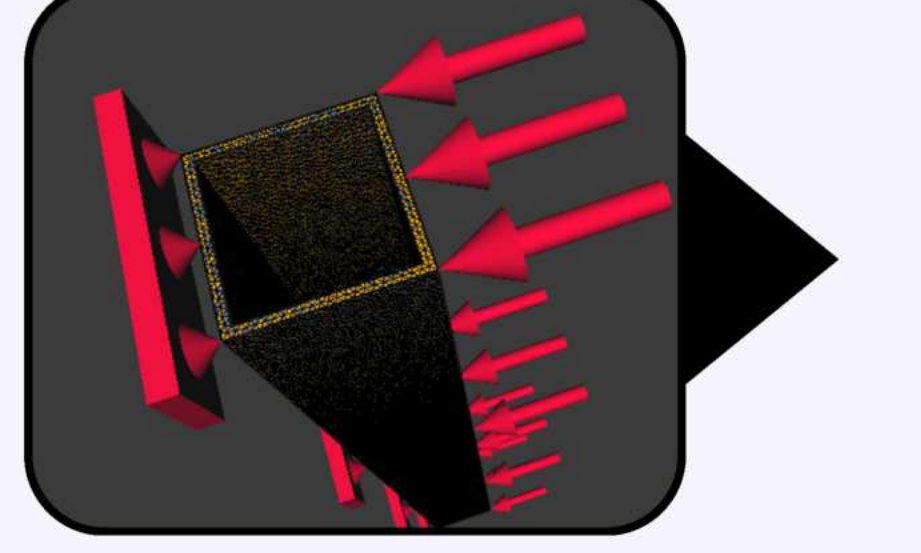
400 K air flow for one second



Structural Stability

Figure created by finalist with FreeCAD

Apply 5 N force on monolith channel



Output major principal stress

Validation: 3D Printing

Results

Compare surface area-to-volume ratios, temperature, and stress distributions for all monolith configurations.

Mass Transfer

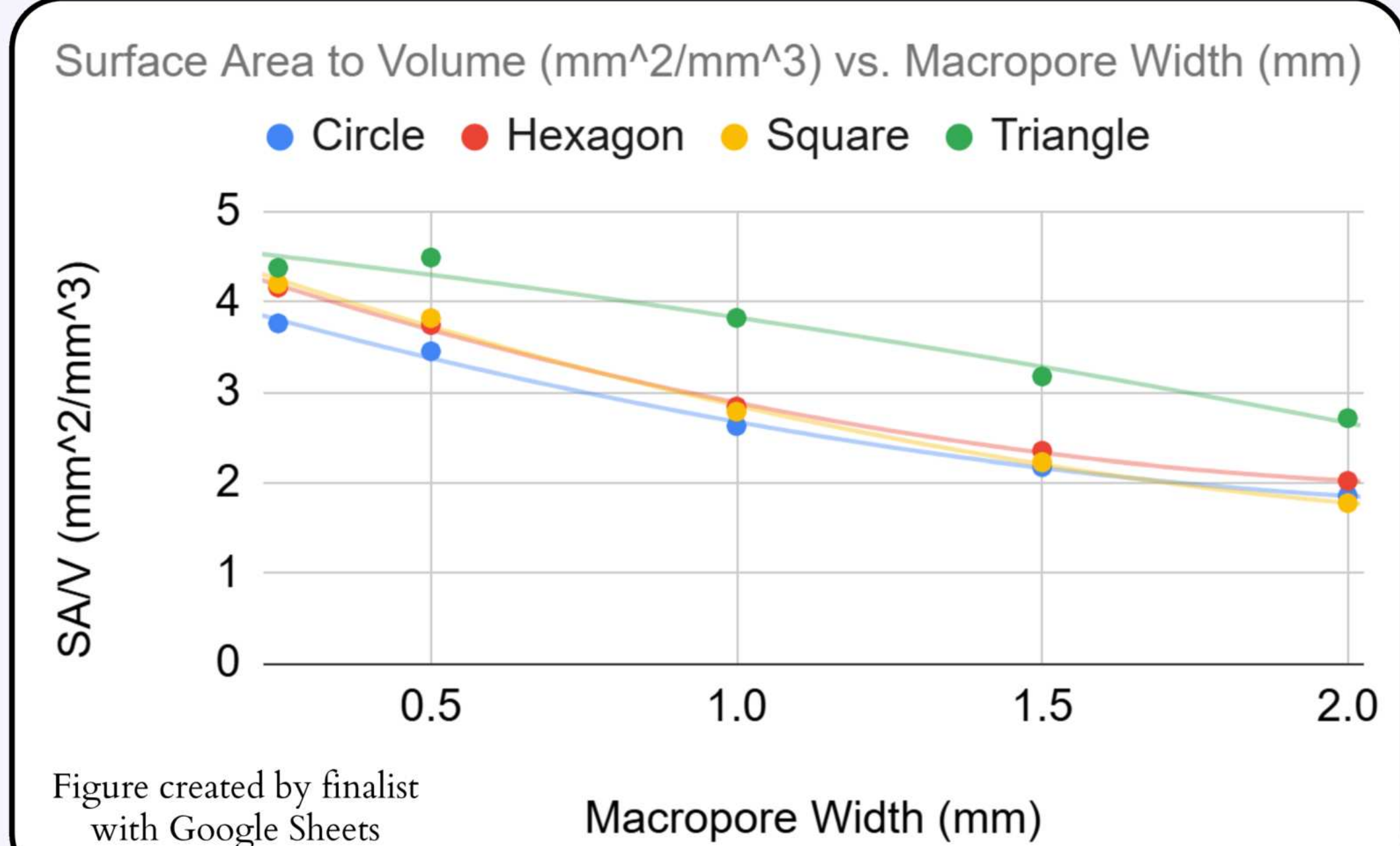


Figure 2 - Surface Area-to-Volume Ratio vs. Pore width/shape

Heat Transfer

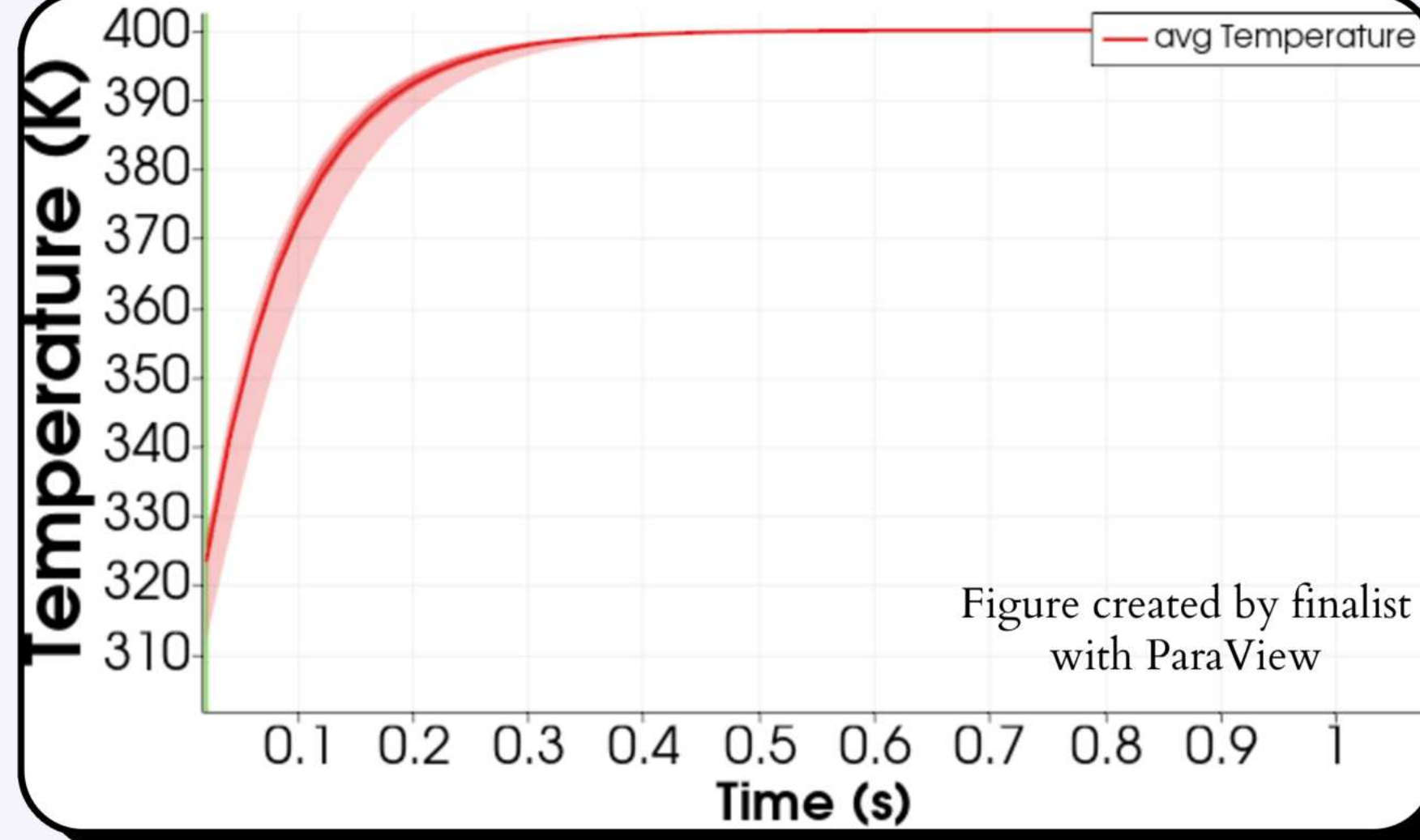


Figure 3 - Temperature vs. Time data for 2 mm and 1.5 mm Triangle-Pore HKUST-1 Monolith

Structural Stability

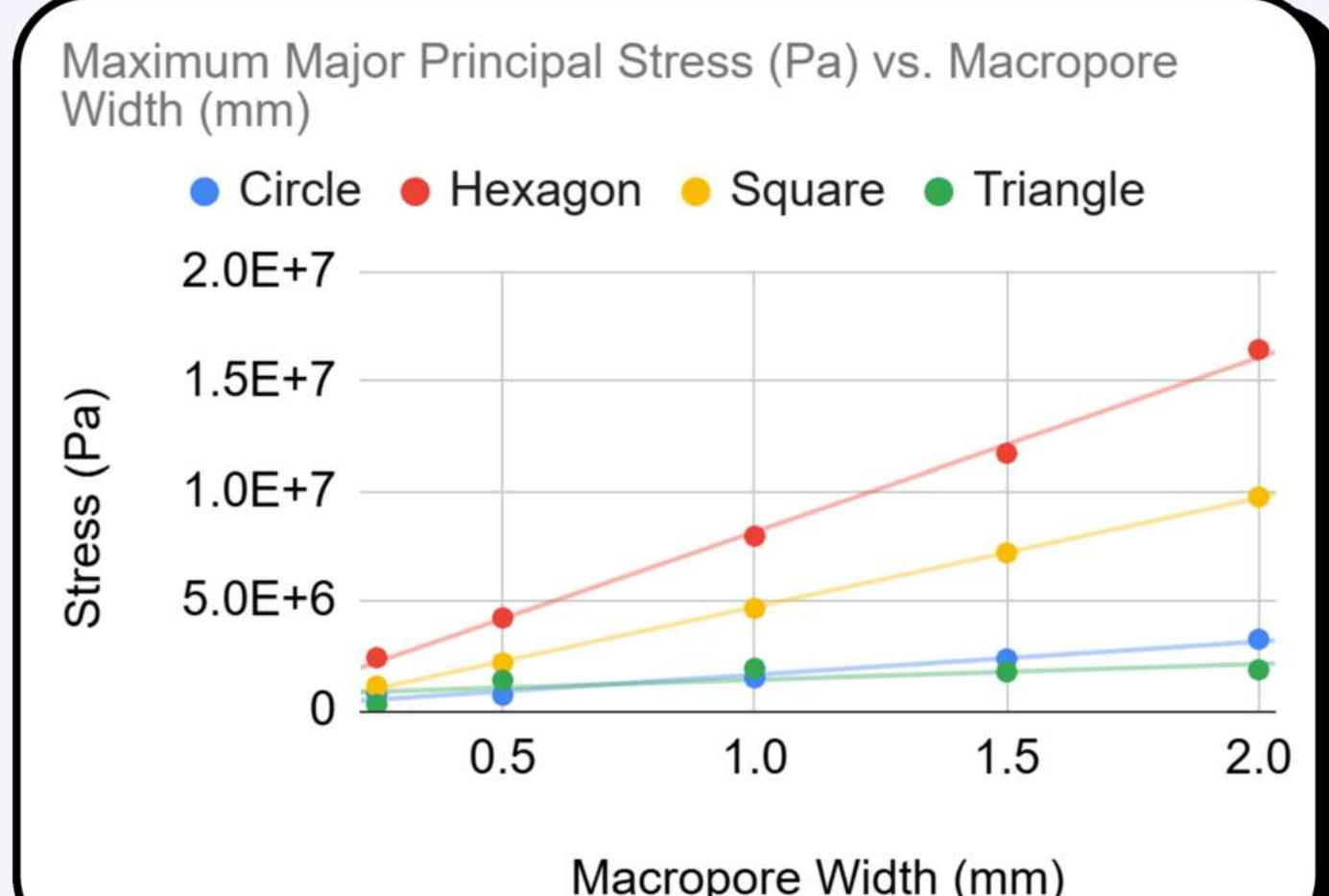


Figure 4 - Maximum Major Principal Stress vs. Macropore Width (HKUST-1)

Material Cost

Table created by finalist with Google Sheets

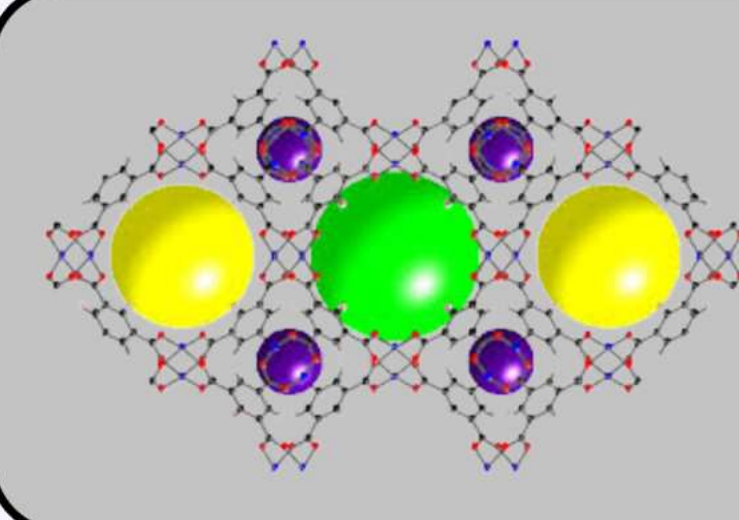
MOF	Material Cost (\$/g)	Bulk Density (g/cm³)	CO2 Adsorption Capacity (mmol/g)	Site Density (mol CO2/cm³)	Volume Cost (\$/cm³)	Adsorption Cost (\$/mol CO2)
HKUST-1	0.054	1.07	5.33	0.0057	0.0578	10.1
MOF-5	0.034	0.22	2.5	0.00055	0.00748	13.6

Table 1 - Adsorption cost calculations for HKUST-1 and MOF-5

Discussion

Findings

Optimal Properties Found



HKUST-1
(Worrall et al., 2016)

MOF Material

Mass Transfer

0.25 mm Triangle Pores



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Heat Transfer

1.5 - 2 mm Triangle Pores

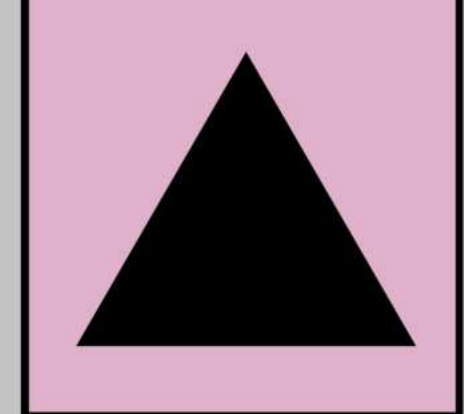


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Structural Stability

0.25 mm Triangle Pores



Figure created by finalist with Canva

Impact

- **Speeds up monolith design** before relying on costly experimental validation
- **Reduces barrier-to-entry** for simulation studies
- **Advances carbon capture technology**, aid in combatting rapid climate change

Future Studies

Custom mass transfer model

Construct HKUST-1 monolith

Physics-informed neural networks

Industry Communication

References

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Stampi-Bombelli, V., Storione, A., Grossmann, Q., & Mazzotti, M. (2024). On comparing packed beds and monoliths for co2 capture from air through experiments, theory, and modeling. *Industrial & Engineering Chemistry Research*, 63(26), 11637–11653. <https://doi.org/10.1021/acs.iecr.4c01392>

Wang, W., Pan, Q., Wang, R., & Ge, T. (2022). Modeling and optimization of a honeycombed adsorbent bed for efficient moisture capture. *Applied Thermal Engineering*, 200, 117717. <https://doi.org/10.1016/j.applthermaleng.2021.117717>

Worrall, S. D., Bissett, M. A., Hill, P. I., Rooney, A. P., Haigh, S. J., Attfield, M. P., & Dryfe, R. A. W. (2016). Metal-organic framework templated electrodeposition of functional gold nanostructures. *Electrochimica Acta*, 222, 361–369. <https://doi.org/10.1016/j.electacta.2016.10.187>